

Test Model on Multiple Datasets

This notebook evaluates a trained model on three test datasets:

- **Natural:** Uniformly random samples from $[1, 10^{13}]$
- **Cheat:** Numbers with prime factors only within the first 100 primes
- **Non-cheat:** Numbers with at least one prime factor outside the first 100 primes

It reports per-class performance for each dataset.

Configuration

Specify the model checkpoint and encoding to test:

```
Testing model: ../models/model_interCRT100_natural/mu/1/checkpoint.pth
Encoding: interCRT100_natural
Task: mu
Results will be saved to: ../test_results/interCRT100_natural_mu
```

Helper Functions

Helper functions loaded!

Check if Model Checkpoint Exists

```
✓ Checkpoint found: ../models/model_interCRT100_natural/mu/1/checkpoint.pth
```

Check Test Data Files

Checking test data files:

```
✓ natural      : ../input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.test (105.83 MB)
✓ cheat        : ../input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.test (105.55 MB)
✓ non_cheat    : ../input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.txt.test (105.82 MB)
```

Run Evaluation on Each Test Dataset

Evaluation function ready!

Evaluate on Natural Dataset

EVALUATING ON NATURAL DATASET

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/checkpoint.pth --eval_data /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --cpu...
```

Full output saved to: ../test_results/interCRT100_natural_mu/eval_natural.log

✓ Evaluation completed successfully!

Overall Accuracy: 51.02%

Evaluate on Cheat Dataset

EVALUATING ON CHEAT DATASET

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/checkpoint.pth --eval_data /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --cpu...
```

Full output saved to: ../test_results/interCRT100_natural_mu/eval_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 41.54%

Evaluate on Non-Cheat Dataset

EVALUATING ON NON-CHEAT DATASET

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/checkpoint.pth --eval_data /mnt/c/Users/ziwen/clair/mobius_case_study/notebooks/./input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --cpu...
```

Full output saved to: ../test_results/interCRT100_natural_mu/eval_non_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 52.19%

Compile Results

✓ Successfully evaluated on 3 dataset(s)

Results saved to: ../test_results/interCRT100_natural_mu/all_results.json

Overall Performance Comparison

OVERALL PERFORMANCE SUMMARY

	Dataset	Accuracy (%)	Acc $\mu=0$ (%)	Acc $\mu=1$ (%)	Acc $\mu=-1$ (%)	Perfect (%)	Correct (%)	XE Loss
0	Natural	51.02	86.632653	57.615670	0.0	51.02	51.02	0.940918
1	Cheat	41.54	44.532090	60.537482	0.0	41.54	41.54	1.721838
2	Non-Cheat	52.19	88.016318	58.336085	0.0	52.19	52.19	0.929030

Summary saved to: ../test_results/interCRT100_natural_mu/overall_summary.csv

Precision, Recall, F1, and Confusion Matrix

Note: The Int2Int evaluator provides per-class accuracy but not the full confusion matrix. To compute precision, recall, and F1 properly, we need to run a custom evaluation that captures predictions vs ground truth.

The following section runs a custom evaluation to extract predictions and compute these metrics.

Classification metrics and parsing functions loaded!

Compute Classification Metrics from Evaluation Results

Use the predictions captured from the evaluation runs above:

```
=====
PARSING PREDICTIONS: NATURAL
=====
Parsing predictions from: ../test_results/interCRT100_natural_mu/eval_dump/debug/vf2m
ludifm/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: CHEAT
=====
Parsing predictions from: ../test_results/interCRT100_natural_mu/eval_dump/debug/xpq8
22nci2/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: NON_CHEAT
=====
Parsing predictions from: ../test_results/interCRT100_natural_mu/eval_dump/debug/xpq8
22nci2/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
Classification metrics computed for 3 dataset(s)
=====
```

Classification Report: Precision, Recall, F1

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CLASSIFICATION REPORT: NATURAL

=====

Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	3049
$\mu=0$	0.58	0.88	0.70	3922
$\mu=1$	0.44	0.58	0.50	3029
accuracy			0.52	10000
macro avg	0.34	0.49	0.40	10000
weighted avg	0.36	0.52	0.43	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.3389	0.3596
Recall	0.4878	0.5219
F1-Score	0.3995	0.4254

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CLASSIFICATION REPORT: CHEAT

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Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	926
$\mu=0$	0.89	0.45	0.59	8367
$\mu=1$	0.07	0.61	0.13	707
accuracy			0.42	10000
macro avg	0.32	0.35	0.24	10000
weighted avg	0.75	0.42	0.51	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.3200	0.7466
Recall	0.3502	0.4154
F1-Score	0.2415	0.5053

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CLASSIFICATION REPORT: NON_CHEAT

Samples evaluated: 10000
Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	926
$\mu=0$	0.89	0.45	0.59	8367
$\mu=1$	0.07	0.61	0.13	707
accuracy			0.42	10000
macro avg	0.32	0.35	0.24	10000
weighted avg	0.75	0.42	0.51	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.3200	0.7466
Recall	0.3502	0.4154
F1-Score	0.2415	0.5053

Confusion Matrices

Figure saved to: ../test_results/interCRT100_natural_mu/confusion_matrices.png

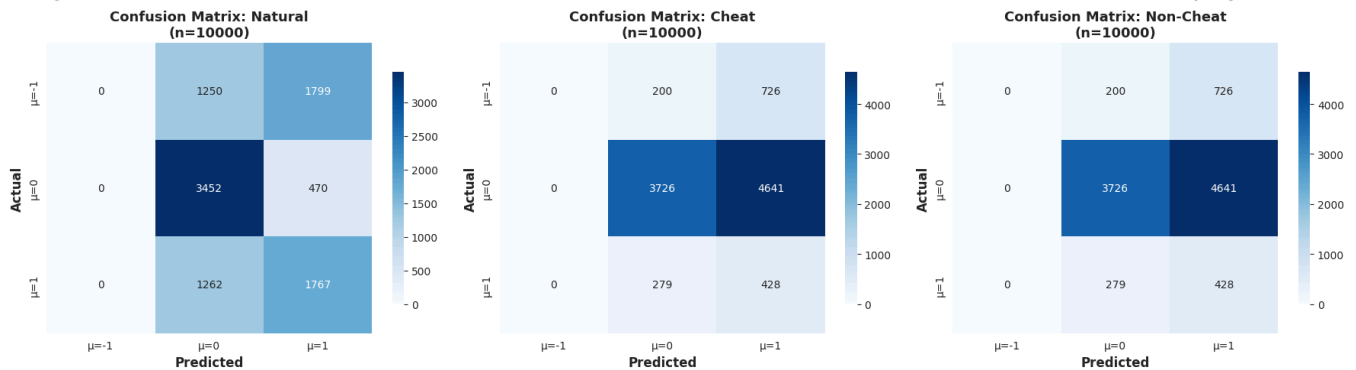
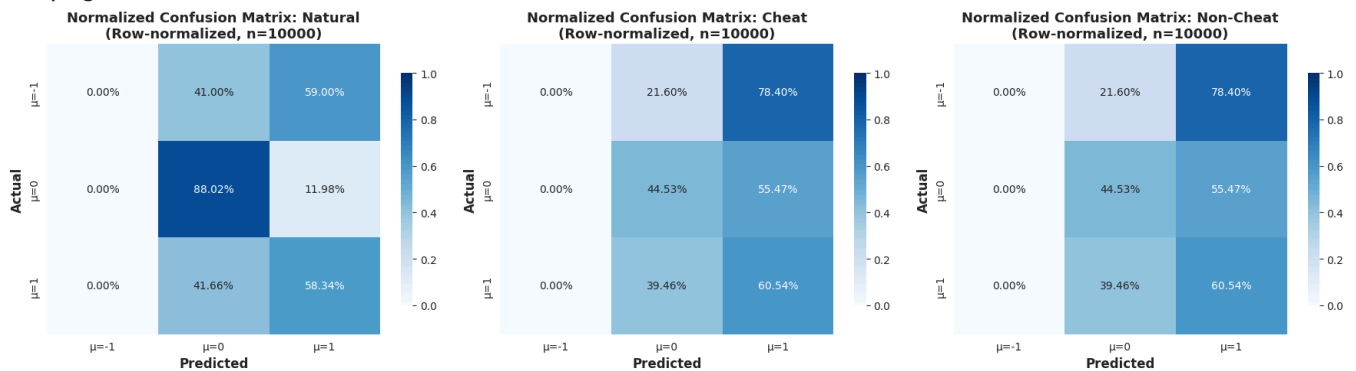


Figure saved to: ../test_results/interCRT100_natural_mu/confusion_matrices_normalize.png



Per-Class Precision, Recall, F1

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PER-CLASS METRICS: NATURAL

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.0000	0.0000	0.0000
1	$\mu=0$	0.5788	0.8802	0.6984
2	$\mu=1$	0.4378	0.5834	0.5002

Saved to: ../test_results/interCRT100_natural_mu/per_class_metrics_natural.csv

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PER-CLASS METRICS: CHEAT

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.0000	0.0000	0.0000
1	$\mu=0$	0.8861	0.4453	0.5927
2	$\mu=1$	0.0739	0.6054	0.1317

Saved to: ../test_results/interCRT100_natural_mu/per_class_metrics_cheat.csv

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PER-CLASS METRICS: NON_CHEAT

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.0000	0.0000	0.0000
1	$\mu=0$	0.8861	0.4453	0.5927
2	$\mu=1$	0.0739	0.6054	0.1317

Saved to: ../test_results/interCRT100_natural_mu/per_class_metrics_non_cheat.csv

Save All Detailed Metrics

All detailed metrics saved to: ../test_results/interCRT100_natural_mu/detailed_classification_metrics.json

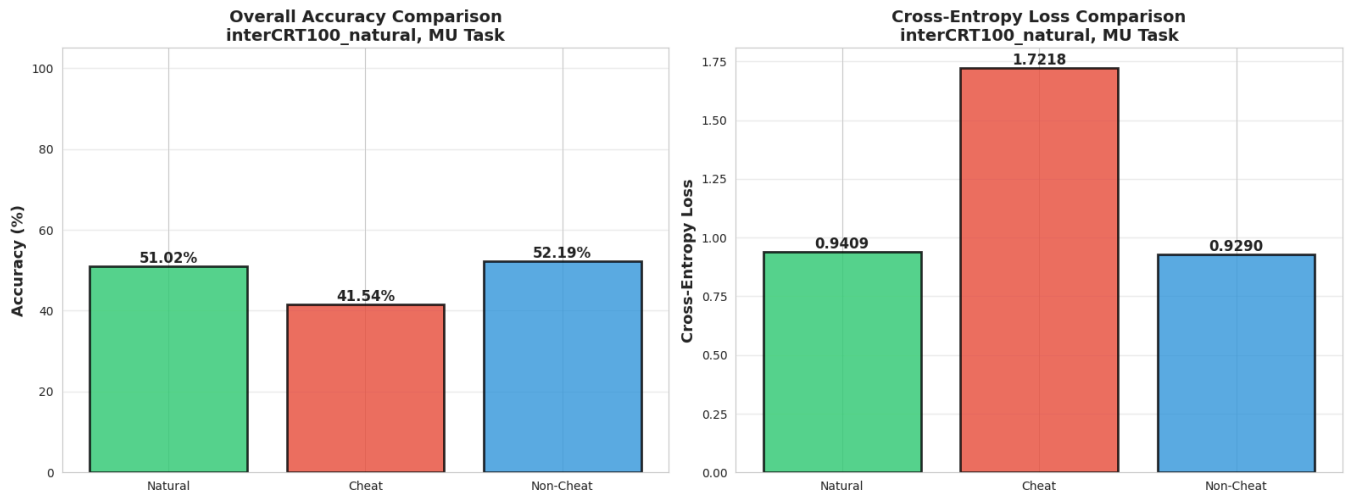
Predictions saved to: ../test_results/interCRT100_natural_mu/predictions_natural.npz

Predictions saved to: ../test_results/interCRT100_natural_mu/predictions_cheat.npz

Predictions saved to: ../test_results/interCRT100_natural_mu/predictions_non_cheat.npz

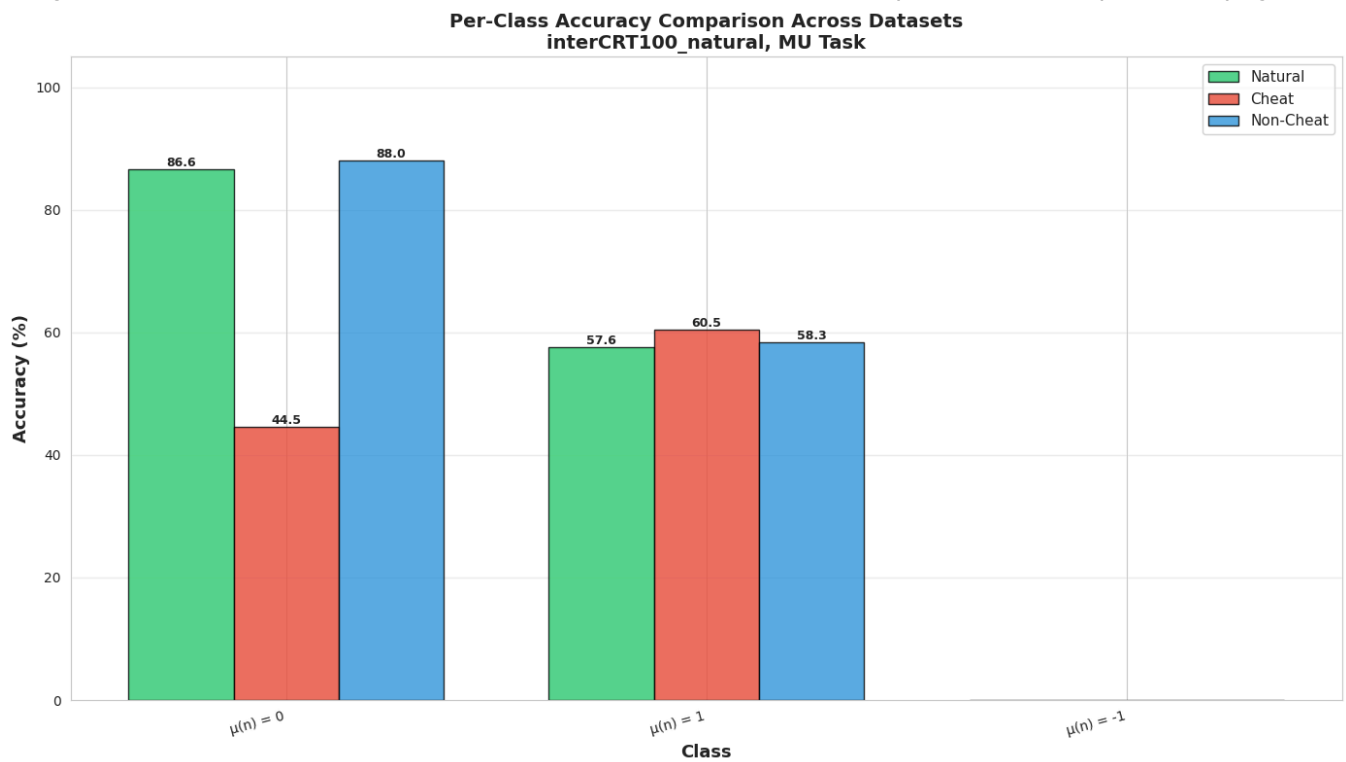
Visualization: Overall Performance Comparison

Figure saved to: ../test_results/interCRT100_natural_mu/overall_comparison.png



Visualization: Per-Class Performance Comparison

Figure saved to: ../test_results/interCRT100_natural_mu/per_class_comparison.png



Manual Inspection: Sample Predictions from Each Test Set

Randomly select 20 samples from each test dataset and display the number, true Mobius value, and model prediction.

```
natural: loaded 10000 numbers
cheat: loaded 10000 numbers
non_cheat: loaded 10000 numbers
```

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SAMPLE PREDICTIONS FROM EACH TEST SET

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DATASET: NATURAL (showing 20 random samples)

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Correct: 10/20 (50.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 290)	1	0	No
1	(sample 549)	0	1	No
2	(sample 1060)	1	0	No
3	(sample 2521)	1	0	No
4	(sample 3349)	-1	0	No
5	(sample 3783)	1	1	Yes
6	(sample 4507)	0	0	Yes
7	(sample 5419)	-1	0	No
8	(sample 5576)	1	1	Yes
9	(sample 6033)	-1	0	No
10	(sample 6267)	-1	0	No
11	(sample 6466)	0	0	Yes
12	(sample 6755)	0	0	Yes
13	(sample 6919)	1	1	Yes
14	(sample 6983)	-1	0	No
15	(sample 7135)	1	1	Yes
16	(sample 7305)	0	0	Yes
17	(sample 8626)	0	0	Yes
18	(sample 9347)	0	0	Yes
19	(sample 9490)	1	0	No

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DATASET: CHEAT (showing 20 random samples)

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Correct: 9/20 (45.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 586)	0	1	No
1	(sample 659)	0	0	Yes
2	(sample 866)	1	1	Yes
3	(sample 1149)	-1	1	No
4	(sample 1198)	0	1	No
5	(sample 1296)	0	0	Yes
6	(sample 1531)	0	1	No
7	(sample 1627)	0	0	Yes
8	(sample 2190)	0	0	Yes
9	(sample 3879)	0	0	Yes
10	(sample 4903)	1	1	Yes
11	(sample 5369)	0	1	No
12	(sample 5756)	0	1	No
13	(sample 6504)	0	1	No
14	(sample 6838)	0	1	No
15	(sample 7332)	0	0	Yes
16	(sample 8818)	0	1	No
17	(sample 8837)	0	1	No
18	(sample 8900)	0	0	Yes
19	(sample 9470)	-1	1	No

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DATASET: NON_CHEAT (showing 20 random samples)

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Correct: 11/20 (55.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 245)	0	0	Yes
1	(sample 1327)	0	0	Yes
2	(sample 1543)	0	0	Yes
3	(sample 1637)	0	1	No
4	(sample 1847)	0	0	Yes
5	(sample 1866)	0	1	No
6	(sample 2124)	0	1	No
7	(sample 2715)	0	1	No
8	(sample 2973)	0	1	No
9	(sample 3016)	0	1	No
10	(sample 5528)	1	1	Yes
11	(sample 6490)	0	0	Yes
12	(sample 6723)	0	0	Yes
13	(sample 6796)	-1	1	No
14	(sample 7186)	0	0	Yes
15	(sample 7756)	-1	1	No
16	(sample 7993)	0	0	Yes
17	(sample 8534)	1	0	No
18	(sample 8981)	0	0	Yes
19	(sample 9566)	0	0	Yes

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